



A Comparison of CF_4 , CHF_3 and $\text{C}_4\text{F}_8 + \text{Ar}/\text{O}_2$ Inductively Coupled Plasmas for Dry Etching Applications

Nomin Lim¹ · Alexander Efremov² · Kwang-Ho Kwon¹

Received: 4 March 2021 / Accepted: 29 June 2021

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2021

Abstract

In this work, we performed the comparative study of plasma parameters, steady-state gas phase compositions and Si reactive-ion etching kinetics in $\text{CF}_4 + \text{O}_2 + \text{Ar}$, $\text{CHF}_3 + \text{O}_2 + \text{Ar}$ and $\text{C}_4\text{F}_8 + \text{O}_2 + \text{Ar}$ gas mixtures with variable O_2/Ar component ratios. It was found that the substitution of Ar for O_2 (a) did not disturb the well-known correlation between the polymerizing ability and the F/C ratio in the original fluorocarbon molecule; (b) causes similar changes in electrons- and ions-related plasma parameters (electron temperature, plasma density, ion bombardment energy); and (c) always suppresses densities of polymerizing radicals and reduces the polymer film thickness. At the same time, the specific effects of oxygen on F atom kinetics result in sufficient differences in their densities and fluxes. It was shown that the dominant etching mechanism for Si in all three gas systems is the chemical etching pathway provided by F atoms (since the contribution of physical sputtering is below 10%) while measured etching rates do not follow the behavior of F atom flux. The phenomenological analysis of heterogeneous process kinetics allowed one to suggest factors influencing the effective reaction probability. These are either the transport of F atoms through thick polymer film (in the case of high-polymerizing $\text{C}_4\text{F}_8 + \text{O}_2 + \text{Ar}$ plasma) or heterogeneous reactions with a participation of oxygen atoms under the condition of thin or non-continuous polymer film (in the case of low-polymerizing $\text{CF}_4 + \text{O}_2 + \text{Ar}$ plasma).

Keywords Fluorocarbon gases · Plasma · Parameters · Active species · Ionization · Dissociation · Etching · Polymerization

Introduction

Fluorocarbon gases with a general formula of $\text{C}_x\text{H}_y\text{F}_z$ are frequently used for the reactive-ion etching (RIE) of silicon and silicon-based materials [1–3]. Among the fluorocarbon gas family, the CF_4 is characterized by the highest z/x ratio and provides the domination of

✉ Kwang-Ho Kwon
kwonkh@korea.ac.kr

¹ Department of Control and Instrumentation Engineering, Korea University, Sejong 30019, Republic of Korea

² Department of Electronic Devices and Materials Technology, State University of Chemistry and Technology, 7 Sheremetevsky av., Ivanovo, Russia 153000

etching over the surface polymerization (in fact, the domination of F atoms flux over that for CF_x radicals) under the typical RIE conditions [4, 5]. This allows one to obtain quite high absolute etching rates, but results in both low SiO_2/Si selectivity and nearly isotropic etching profiles for Si [2, 4]. The last effect is due to the spontaneous chemical reaction between silicon and fluorine atoms [5]. Oppositely, more polymerizing fluorocarbons with $z/x < 3$ (in particular, CHF_3 and C_4F_8) provide much higher etching anisotropy for Si (due to the passivation of side walls by the fluorocarbon polymer film [3–5]) together with the much higher etching selectivity over the SiO_2 (due to the much lower polymer film thickness on the oxygen-containing surface [4, 5]). Accordingly, the negative consequences are the loss in absolute etching rates and sufficient etching residues. That is why all fluorocarbons are frequently combined with Ar or O_2 in order to accelerate the physical etching pathway, increase the F atoms yield and suppress surface polymerization [5, 6]. Therefore, the choice of an appropriate fluorocarbon gas, additive components and their mixing ratios is a powerful tool to optimize output etching characteristics (etching rate, anisotropy and selectivity in respect to mask and/or under-layer material) for a given type of treated material. This task cannot be solved without understanding of relationships between processing conditions, internal plasma parameters, steady-state densities of plasma active species and heterogeneous process kinetics.

Until now, there were many works dealt with investigations of plasma parameters and gas-phase compositions in $\text{CF}_4 + \text{Ar}/\text{O}_2$ [7–10, 12], $\text{CHF}_3 + \text{Ar}/\text{O}_2$ [11–13] and $\text{C}_4\text{F}_8 + \text{Ar}/\text{O}_2$ [9, 14–17] plasmas. In fact, results of these researches allowed one (a) to figure out key gas-phase processes determining kinetics of fluorine atoms and polymerizing radicals; (b) to compose kinetic schemes (sets of reactions with corresponding rate coefficients) for the adequate description of plasma chemistry in the presence of oxygen and argon; and (c) to understand basic responses of both etching and polymerization kinetics to changes in processing parameters (input power, pressure and gas mixing ratios). In addition, a series experimental studies (for example, Refs. [18–23] as well as earlier ones summarized in monographs [4–6]) brought a decent knowledge on etching kinetics and mechanisms for Si, SiO_2 and Si_3N_4 in various fluorocarbon gas plasmas, including mixtures with Ar and O_2 . Corresponding results provide the ability to understand the basic interconnections between etching rates and processing conditions in polymerizing plasmas as well as to associate features of etching selectivities and profiles with the type of fluorocarbon gas used for the etching process. At the same time, when analyzing existing data, one can formulate at least two principal problems that require additional research efforts. First of all, the works reported on parameters and compositions for CF_4 , CHF_4 and C_4F_8 —based plasmas, including mixtures of these gases with oxygen, often used different processing conditions (input parameters and/or reactor types and geometries) as well as discussed plasma chemistry with different approaches and reaction schemes. Such situation makes it impossible to compare $\text{CF}_4 + \text{Ar}/\text{O}_2$, $\text{CHF}_3 + \text{Ar}/\text{O}_2$ and $\text{C}_4\text{F}_8 + \text{Ar}/\text{O}_2$ plasmas in respect to kinetics of plasma active species, gas-phase plasma characteristics influencing various heterogeneous interaction pathways and reactive-ion etching performances for the given treated material. The second problem came from a purely experimental nature of many works dealt with both etching and polymerization kinetics in fluorocarbon gas plasmas. In fact, most of authors provided only the phenomenological study how operating conditions do influence processing result as well as discussed formal correlations between polymer film thicknesses/composition, etching rates and etching selectivities for oxygen-containing and oxygen-free surfaces. As such, even very detailed and accurate studies (for example, Refs. [18–21]) reported nothing about gas-phase-related mechanisms which transfer the change in operating conditions to heterogeneous process kinetics. Though Matsui et al. [19] have made a decent attempt to analyze the effect of radical flux in $\text{C}_4\text{F}_8 + \text{Ar}/\text{O}_2$ plasma by varying the

C_4F_8 flow rate at constant electron density and ion bombardment energy, their experimental approach is quite far from the real reactive-ion etching conditions. The questionable thing is also that they attributed differences in Si, SiO_2 and Si_3N_4 etching rates only to polymer film thickness without accounting for both sputtering yields and reaction probabilities for F atoms. In our opinion, poorly studied relationships between gas-phase and heterogeneous chemistries limit the significance of corresponding results (as latters are surely valid only for an investigated range of processing conditions) and causes uncertainties in interpretations of etching mechanisms. Such situation retards both optimization of reactive-ion etching technologies and the overall progress in the electronic device fabrication field.

Our previous works [9, 10, 16, 17, 24–26] suggested an advanced research scheme as a combination of etching experiments, plasma diagnostics by Langmuir probes and plasma modeling. Such a method clearly demonstrates how processing parameters do effect on gas-phase plasma characteristics as well as allows one to analyze etching mechanisms with model-predicted fluxes of plasma active species. Particularly, Ref. [9] provided the comparison of $CF_4 + O_2 + Ar$ and $C_4F_8 + O_2 + Ar$ plasmas in terms of ions-related parameters, electron gas characteristics and formation/decay kinetics for neutral species. It was found that (a) the presence of O_2 does not reduce the difference in polymerizing abilities for CF_4 - and C_4F_8 -based plasmas; and (b) the variation of O_2/Ar mixing ratio causes opposite effects on densities of fluorine atoms. In addition, Ref. [26] studied plasma chemistry in oxygen-free $CF_4 + Ar$, $CHF_3 + Ar$ and $C_4F_8 + Ar$ gas systems under one and the same operating conditions. It was demonstrated that evident dissimilarities in both ion and neutral fluxes reasonably follow from corresponding differences in plasma parameters and gas-phase reaction mechanisms. Unfortunately, the common feature of these works is that the analysis of etching and polymerization kinetics was made in a suggestive scale, without real etching experiments. Though related conclusions showed the not bad qualitative agreement with earlier published data, the direct confirmation is still expected. The idea of this work was to apply the full (i.e. with etching experiments) research scheme for the comparative study of $CF_4 + O_2 + Ar$, $CHF_3 + O_2 + Ar$ and $C_4F_8 + O_2 + Ar$ inductively coupled plasmas. The choice of fluorocarbon components was motivated by the continuous decrease of z/x ratio in the sequence of CF_4 – CHF_3 – C_4F_8 . As was demonstrated in Refs. [20, 26], corresponding non-oxygenated plasmas provide sufficient differences in their polymerizing abilities and fluorine atom densities. Accordingly, main goals of this study were (1) to compare how the change in O_2/Ar mixing ratio does influence electrons- and ions-related plasma parameters; (2) to analyze differences in densities of fluorine atoms and polymerizing radicals in the presence of oxygen; (3) to suggest features of etching and polymerization kinetics based on gas-phase plasma characteristics; and (4) to verify above suggestions through the comparison of Si etching kinetics and mechanisms in given gas systems. The silicon was selected as the test material because of the well-known etching mechanism provided by the spontaneous reaction with F atoms. In our opinion, this allows one the better understanding of side factors influencing the chemical etching pathway in various gas systems.

Experimental and Modeling Details

Experimental Setup and Procedures

Experiments were performed in the planar (with the top-side flat coil) inductively coupled plasma (ICP) reactor, the same as that used in our previous works [24–26]. Plasma

was excited using the 13.56 MHz power supply while another 12.56 MHz rf generator biased the bottom electrode in order to control the negative dc bias voltage ($-U_{dc}$). The latter was measured using the high-voltage probe (AMN-CTR, Youngs In Eng.). All experimental series were carried out at fixed set of processing parameters: gas pressure ($p=6$ mTorr), total gas flow rate ($q=40$ sccm), input power ($W=700$ W) and bias power ($W_{dc}=200$ W). The variable parameter was the O_2/Ar mixing ratio in CF_4+O_2+Ar , CHF_3+O_2+Ar and $C_4F_8+O_2+Ar$ gas mixtures. In each gas system, the initial composition of feed gas was adjusted by partial flow rates for individual mixture components. Particularly, the flow rate of CF_4 , CHF_3 or C_4F_8 was always 20 sccm that provided the constant 50% fraction of fluorocarbon component. The change in q_{O_2} in the range of 0–20 sccm was accompanied by the proportional decrease in q_{Ar} in order to maintain $q_{O_2}+q_{Ar}=\text{const}$. As such, an increase in $y_{O_2}=q_{O_2}/q$ from 0 to 50% corresponded to the full substitution of Ar for O_2 .

Plasma diagnostics by the double Langmuir probe (DLP2000, Plasmart Inc.) delivered data on electro-physical plasma parameters, such as electron temperature (T_e) and ion current density (J_+). Latters were extracted from measured current–voltage (I–V) curves using the well-known statements of double Langmuir probe theory [27]. In order to minimize experimental errors due to the polymerization on probe tips, these were treated in 50% Ar+50% O_2 plasma before and after each measurement. Our previous works [16, 25, 26] have demonstrated the efficiency of such cleaning procedure to provide adequate diagnostics results in high-polymerizing fluorocarbon gas plasmas.

Silicon etching kinetics was studied using fragments of Si (111) wafer with an average size of $\sim 2\times 2$ cm. Samples were placed in the middle part of the bottom electrode. The bottom electrode had a built-in water-flow cooling system that allowed one to maintain its temperature (T_s) at the nearly constant value within plasma existence times up to 5 min. The small sample size allowed one to neglect the loading effect as well as to provide the etching regime controlled by heterogeneous process kinetics. Preliminary experiments showed no principal differences in both raw I–V curves and related plasma parameters obtained with and without sample loading. Therefore, one can neglect the sensitivity of gas-phase plasma parameters to etching products as well as consider the gas phase to be the permanent source of active species. Etched depths (Δh) were determined using the surface profiler (Alpha-Step 500, Tencor) for the processing time $\tau=1$ min. For this purpose, we developed a partial surface masking by the photoresist AZ1512 with a thickness of ~ 1.5 μm . Quasi-linear shapes of $\Delta h=f(\tau)$ curves in all three gas mixtures allowed one to assume the steady-state etching kinetics as well as pointed out on the very weak change in the sample temperature within the given processing time. The latter is because the continuously increasing sample temperature is expected to produce Arrhenius-like growth of Δh due to the spontaneous reaction mechanism between Si and F atoms. Based on above suggestions, we calculated Si etching rates as $R=\Delta h/\tau$ as well as did not account for sample temperature-related effects when analyzing changes in etching kinetics with increasing O_2 fraction in a feed gas.

Approaches for Analysis of Gas-Phase Kinetics

In order to analyze the influence Ar/O_2 mixing ratio on kinetics and densities of plasma active species, we used a simplified 0-dimensional (global) kinetic model [9, 10]. Kinetic schemes (sets of chemical reactions with corresponding rate coefficients)

were taken from our previous works that dealt with the modeling of $\text{CF}_4 + \text{Ar}/\text{O}_2$ [9, 10], $\text{CHF}_3 + \text{Ar}/\text{O}_2$ [12, 13] and $\text{C}_4\text{F}_8 + \text{Ar}/\text{O}_2$ [9, 17] plasmas. Basic model approaches were as follows:

1. The electron energy distribution function (EEDF) has the nearly Maxwellian shape. Such situation results from the quite high ionization degree ($n_+/N > 10^{-4}$ where $N = p/k_B T_{\text{gas}}$ is the total gas density) that pre-determines the sufficient contribution of equilibrium electron–electron collisions to the overall electron energy balance. Accordingly, several works [7, 8, 11, 14, 15] have demonstrated the applicability of Maxwellian EEDFs to describe electron-impact kinetics in low-pressure CF_4 -, CHF_3 - and C_4F_8 -based plasmas. This allowed one to obtain the electron-impact rate coefficients in a form of fitting expressions $k = f(T_e)$ [8, 11, 15].
2. The condition $p, W = \text{const}$ always provides the nearly constant gas temperature, T_{gas} . The indirect proof for this suggestion is the nearly constant temperature of external chamber wall obtained for different gas mixtures and Ar/O_2 mixing ratios at fixed plasma on time. Similarly to our previous works [13, 16, 17], we assumed $T_{\text{gas}} \sim 600$ K. In fact, this represents the typical value for given processing conditions, reactor type and geometry. This allowed one to operate with the constant rate coefficients for gas-phase atom-molecular reactions.
3. The heterogeneous loss of atoms and radicals follow the first-order recombination kinetics with known recombination probabilities [14, 28]. We assumed also that all recombination probabilities are independent on both type of fluorocarbon component and Ar/O_2 mixing ratio. The reason is the stable chamber walls conditions because of (a) no re-deposition of reaction products due to the low sample size; and (b) the nearly constant internal wall temperature. The indirect proof for the last suggestion is the nearly constant external wall temperature, as was mentioned above.
4. The electronegativity of low pressure CF_4 , CHF_3 and C_4F_8 plasmas is low enough to assume $n_e \approx n_+$, where n_e is the electron density, and n_+ is the total density of positive ions. The reasonability of such assumption at $p < 20$ mTorr was surely demonstrated in Refs. [8, 11, 28]. In addition, Lee et al. [29] reported $n_-/n_e < 0.1$ and $n_e \approx n_+$ in pure O_2 plasma at $p = 10$ mTorr and comparable with our case input power density. This allows one to obtain the total density of positive ions from measured J_+ without accounting for the influence of negative ions on the ion Bohm velocity:

$$n_+ \approx \frac{J_+}{0.61e\sqrt{eT_e/m_i}} \quad (1)$$

The effective ion mass was evaluated as $m_i = \left(\sum y_{X_i^+}/m_{X_i^+}\right)^{-1}$, where $y_{X_i^+}$ and $m_{X_i^+}$ are partial ion fractions and masses, respectively. For each type of positive ion, it was suggested that $y_{X^+} \sim k_{iz}y_X/\sqrt{1/m_{X^+}}$ [19], where y_X is the fraction of corresponding neutral particle with the ionization rate coefficient of k_{iz} [14, 28, 30]. It should be noted that the combination of used modeling algorithm and kinetic scheme provides the adequate description of plasma chemistry in given gas systems [10, 13, 17]. This follows from both general reasonability of model-predicted data and their agreements with experiments, as will be mentioned below.

Approaches for Analysis of Etching Kinetics

The basic features of reactive-ion etching processes in fluorocarbon-based plasmas as well as related reaction mechanisms for Si, SiO₂ and Si₃N₄ have been discussed in detail in Refs. [18–23]. When summarizing these findings, we have suggested a set of phenomenological approaches for the analysis of reactive-ion etching kinetics [16, 17, 25, 26]. These are as follows:

1. Under the reactive-ion etching conditions (i.e. when the ion bombardment energy exceeds the sputtering threshold for the target surface), the measured etching rate, R , may be represented in a form of two summands, $R_{phys} + R_{chem}$. These represent physical (sputtering by ion bombardment) and chemical (ion-assisted chemical reaction) etching pathways, respectively.
2. The ion-assisted chemical reaction has the rate of $\gamma_R \Gamma_F$, where γ_R is the effective reaction probability which accounts for the net effect from all heterogeneous stages, and $\Gamma_F \approx n_F v_T / 4$ is the fluorine atom flux. In general case, the effective reaction probability $\gamma_R \approx s_0(1 - \theta)$ depends on surface temperature (through the sticking coefficient s_0) as well as on any factor influencing the fraction of free adsorption sites for F atoms ($1 - \theta$). In polymerizing plasmas, γ_R decreases with increasing thickness of the fluorocarbon polymer film when the latter becomes to be enough to provide $\Gamma'_F / \Gamma_F < 1$, where Γ'_F is the flux of F atoms on the polymer film/etched surface interface.
3. The physical sputtering (as well as any other ion-related etching effect with a purely physical nature: breaking of chemical bonds between surface atoms, ion-stimulated desorption of reaction products) has the rate of $Y_S \Gamma_+$, where $Y_S \sim \sqrt{M_i \varepsilon_i}$ is the sputtering yield, $M_i = m_i N_A$ is the effective ion molar mass, $\varepsilon_i = |-U_f - U_{dc}|$ is the ion bombardment energy, $-U_f \approx 0.5 T_e \ln(m_i / 2.3 m_e)$ is the floating potential, and $\Gamma_+ \approx J_+ / e$ is the flux of positive ions. Accordingly, the relative change in the corresponding process rate may be traced by the parameter $\sqrt{M_i \varepsilon_i} \Gamma_+$, or by $\sqrt{\varepsilon_i} \Gamma_+$ under the condition of $M_i \approx \text{const}$.
4. The growth of the fluorocarbon polymer film is provided by the CH_xF_y ($x + y < 3$) radicals, and the polymerization ability increases in fluorine-poor plasmas. Accordingly, the polymer deposition rate may be traced by the Γ_{pol} / Γ_F ratio, where Γ_{pol} is the total flux of polymerizing radicals while parameters $\Gamma_{pol} / \sqrt{M_i \varepsilon_i} \Gamma_+ \Gamma_F$ and $\Gamma_{pol} / \Gamma_O \Gamma_F$ characterize the change in the polymer film thickness due to physical (destruction by ion bombardment) and chemical (etching by O atoms) destruction pathways, respectively.

Results and Discussion

Differences in electrons- and ions-related plasma parameters for CF₄ + Ar, CHF₃ + Ar and C₄F₈ + Ar plasmas were the subject of detailed discussion in Refs. [12, 26]. That is why below we will focus the attention only on effects related to the addition of oxygen. From Fig. 1, it can be understood that the substitution of Ar for O₂ in all three gas mixtures causes the qualitatively similar deformation of EEDF as well as exhibits no principal differences in the formation/decay balance for charged species. Corresponding data may be briefly commented as follows:

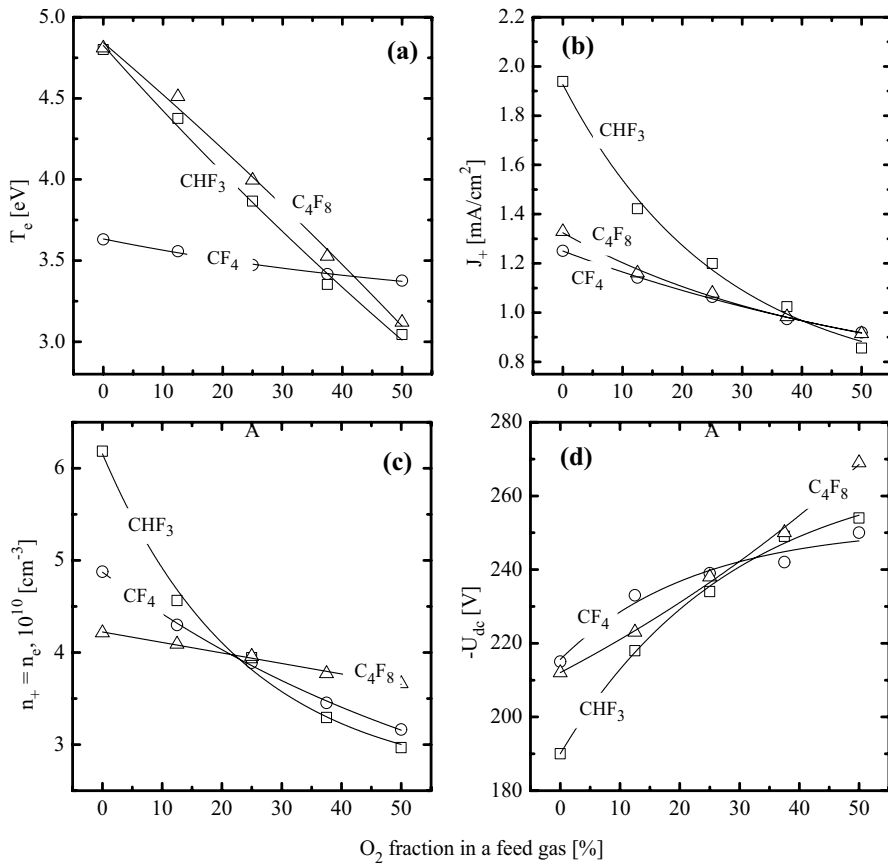


Fig. 1 Electrons- and ions-related plasma parameter as functions of O_2 content in a feed gas at $p=6$ mTorr, $W=700$ W and $W_{dc}=200$ W: **a** electron temperature; **b** ion current density; **c** densities of charged species; and **c** negative dc bias

- A decrease in T_e (Fig. 1a) results from an increase in electron energy losses due to low-threshold excitations for O_2 and other molecular species which appear in a gas phase as products of plasma chemical reactions—CO, CO_2 , FO, CFO and CF_2O [9, 10, 13]. In particular, the first excitation potential for Ar atom is ~ 11.6 eV while O_2 is characterized by the almost continuous energy loss spectrum starting from ~ 0.2 eV. The last feature is provided by the vibration excitation R1 ($\epsilon_1 = 0.16$ eV) as well as by the formation of metastable states in R2 ($\epsilon_2 = 0.98$ eV) and R3 ($\epsilon_3 = 1.64$ eV), where ϵ is the corresponding threshold energy:





One can reasonably expect that similar differences in excitation threshold energies do exist also between Ar and any other molecular component that appears with the addition of O_2 . As such, since the substitution of Ar for O_2 increases the fraction of molecular species in a gas phase [9, 13], it always reduces the amount of high-energy electrons in the EEDF and shifts the mean electron energy toward lower values. A bit weaker effect for $\text{CF}_4 + \text{O}_2 + \text{Ar}$ plasma is because of the lower electron temperature in CF_4 itself. The last feature is probably due to higher electron energy losses in collisions with CF_4 molecules compared with those for dominant components of CHF_3 - and C_4F_8 -based plasmas (Fig. 2). The possible reason is that the bigger-sized and multi-atomic CF_4 provides more vibrational and electronic excitation pathways with higher cross-sections than those for HF and CF_2 .

- A decrease in J_+ toward O_2 -rich plasmas (Fig. 1b) reflects the change in densities of charged species (Fig. 1c). The main mechanism of this phenomenon is that the decreasing T_e suppresses ionization through decreasing ionization rate coefficients, k_{iz} , for all types of neutral species. The high sensitivity of k_{iz} to the change in electron temperature is provided by the condition $\varepsilon_{iz} \approx 12\text{--}15 \text{ eV} > (3/2)T_e$ [31], where ε_{iz} is the threshold energy for ionization, and $(3/2)T_e$ is the mean electron energy in the case of Maxwellian EEDF [5]. In addition, an increase in y_{O_2} results in increasing densities of electronegative species due to both O_2 itself and oxygen-containing reaction products. This accelerates the decay rates for positive ions and electrons through the ion-ion recombination and dissociative attachment, respectively.
- An increase in $-U_{dc}$ (Fig. 1d) is mainly due to the opposite change in ion flux $\Gamma_+ \approx J_+/e$. The reason is that the lower Γ_+ leads to the weaker compensation for the excess negative charge provided by bias power source at $W_{dc} = \text{const}$.

When analyzing the chemistry of neutral species, one can conclude that given gas systems exhibit sufficient differences in electron-impact dissociation mechanisms for original fluorocarbon molecules as well as provide specific interaction pathways of first-step dissociation products with other species, including O and O_2 . Since basic gas-phase reaction schemes for CF_4 -, CHF_3 - and C_4F_8 -based plasmas have been analyzed in our previous works [10, 13, 17], the goal of this work is summarize most principal issues determining differences in densities of F atoms and polymerizing radicals under one and the same operating conditions.

In the $\text{CF}_4 + \text{Ar}$ plasma, dominant fluorine-containing components are original CF_4 molecules, CF_3 radicals and F itself [7, 8, 32] (Fig. 2a). The formation of F atoms is mainly provided by R4 and R5 for $x=3, 4$. The decay of atomic species is due to their heterogeneous recombination in R6 and R7 while corresponding gas-phase counterparts (mainly R7 for $x=3$) bring only $\sim 10\%$ to the total F atom loss rate:



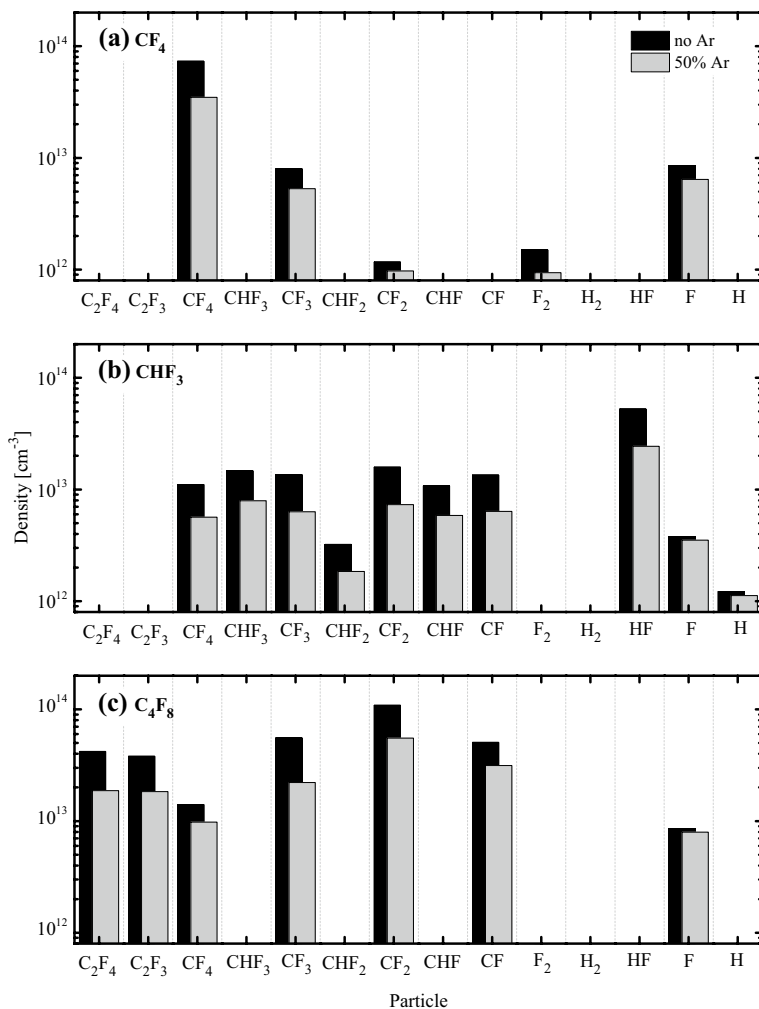


Fig. 2 Steady-state densities of neutral species in non-oxygenated CF₄- (a), CHF₃- (b) and C₄F₈- (c) based plasmas. Black bars are for pure fluorocarbon gas while gray ones relate to their equal mixing with Ar. Processing conditions correspond to Fig. 1



The substitution of Ar for O₂ in the CF₄ + O₂ + Ar gas mixture rapidly reduces densities of CF_x radicals (by ~100 times for x=1, ~400 times for x=2 and ~100 times for x=3, see Fig. 3(b)) through their decomposition in R8 with both ground-state O(³P) (denoted as O) and metastable O(¹D) (denoted as O*) oxygen atoms ($k_8 \sim 6.1 \times 10^{-11}$ cm³/s for x=1 and $\sim 3.2 \times 10^{-11}$ cm³/s for x=2, 3) as well as introduces new formation pathways for F atoms. In addition to R8, most important ones are electron-impact reactions R9 for x=1, 2 and R10:

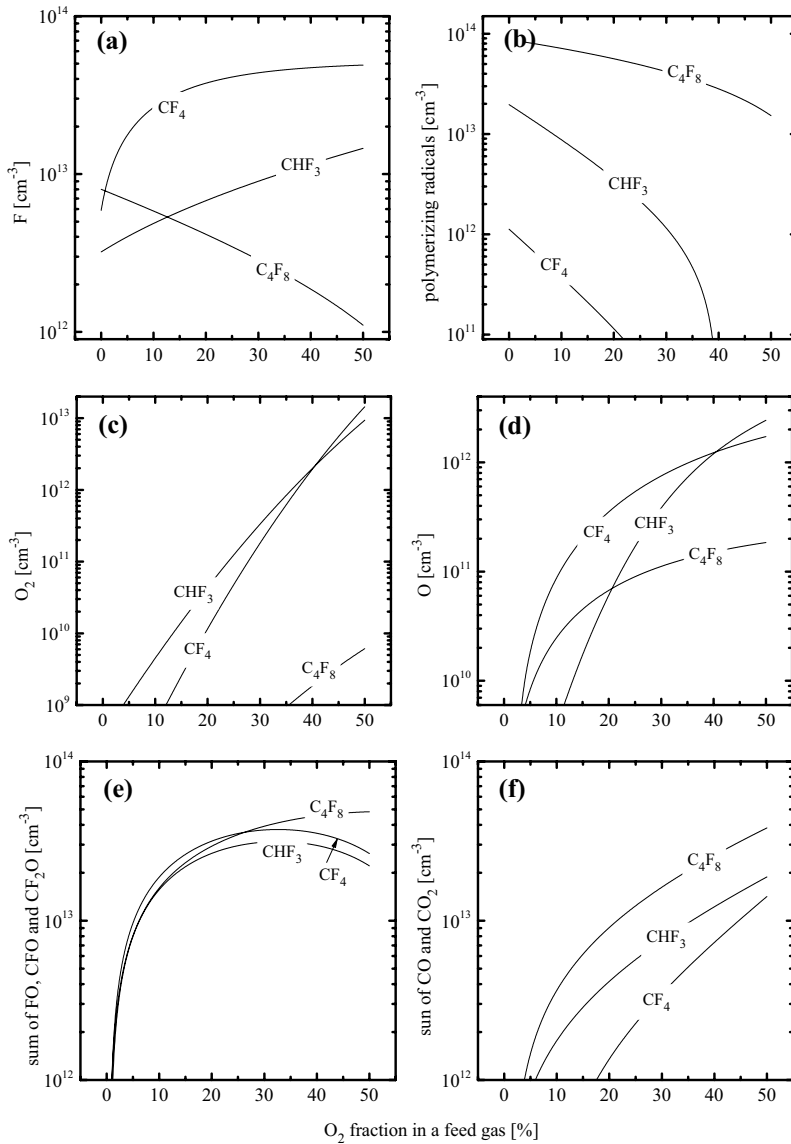
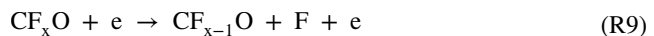
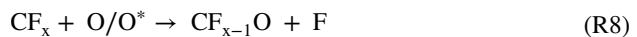
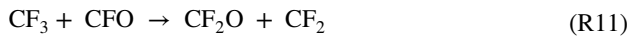


Fig. 3 Steady-state densities of selected neutral species as functions of O_2 content in a feed gas: **a** F atoms; **b** total density of polymerizing radicals ($n_{CF} + n_{CF_2}$ for CF_4 - and C_4F_8 -based plasmas while $n_{CF} + n_{CHF} + n_{CF_2}$ for CHF_3 -based plasma); **c** O_2 molecules; **d** O atoms; **e** total density of FO and CF_xO species; and **f** total density of CO_x species. Processing conditions correspond to Fig. 1





As can be seen from Fig. 3(d), the total density of source species for R9 and R10 exhibits a sufficient (by more than an order of magnitude for 5–30% O_2) increase toward O_2 -rich plasmas. In particular, the fast formation of CF_2O is provided by R11 ($k_{11} \sim 1.1 \times 10^{-11} \text{ cm}^3/\text{s}$), R12 ($k_{12} \sim 1.0 \times 10^{-11} \text{ cm}^3/\text{s}$) and R13 ($k_{13} \sim 8.0 \times 10^{-11} \text{ cm}^3/\text{s}$) while similar effects for CFO and FO are due to R9, R14 ($k_{14} \sim 3.1 \times 10^{-11} \text{ cm}^3/\text{s}$) and the heterogeneous process R15:

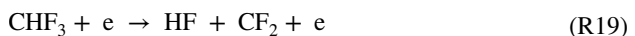


Except these, the condition of $y_{\text{O}_2} > y_{\text{Ar}}$ noticeably rises rates of atom-molecular reactions R16: ($k_{16} \sim 2.5 \times 10^{-11} \text{ cm}^3/\text{s}$ and $\sim 5.0 \times 10^{-11} \text{ cm}^3/\text{s}$ for ground-state and metastable oxygen atoms, respectively), and R18 ($k_{18} \sim 1.0 \times 10^{-10} \text{ cm}^3/\text{s}$):



that also support the F atom formation kinetics. As a result, the substitution of Ar for O_2 leads to a continuous increase in the F atom formation rate [9, 10, 24] and thus, the F atom density (Fig. 3a). It is important to note also that the opposite change in n_{F} and $n_{\text{CF}} + n_{\text{CF}_2}$, as can be seen from Fig. 3a, b, creates a favorable condition for etching, but not for polymerization. This can be clearly seen from Fig. 4a which indicates a decrease in the $\Gamma_{\text{pol}}/\Gamma_{\text{F}}$ ratio.

In the $\text{CHF}_3 + \text{Ar}$ plasma, main fluorine-containing species are HF, CHF_3 and CF_x ($x = 1-3$) (Fig. 2b) [12, 13]. The high density of HF (that is in good agreement with previously published data [33, 34]) is provided by two mechanisms, such as a) the direct formation of these species in R19; and 2) the high efficiency of gas-phase reactions R20 ($k_{20} \sim 3.3 \times 10^{-11} \text{ cm}^3/\text{s}$ for $x = 1$ and 2 while $\sim 1.6 \times 10^{-13}$ for $x = 3$), R21 ($k_{21} \sim 3.1 \times 10^{-10} \text{ cm}^3/\text{s}$ for $x = 1$ and 2) and R22 ($k_{22} \sim 1.2 \times 10^{-11} \text{ cm}^3/\text{s}$ for $x = 1$, $\sim 2.2 \times 10^{-11} \text{ cm}^3/\text{s}$ for $x = 2$ and $\sim 7.9 \times 10^{-11} \text{ cm}^3/\text{s}$ for $x = 3$). Accordingly, main formation channels for F atoms are R23 and R5 for $x = 2, 3$:



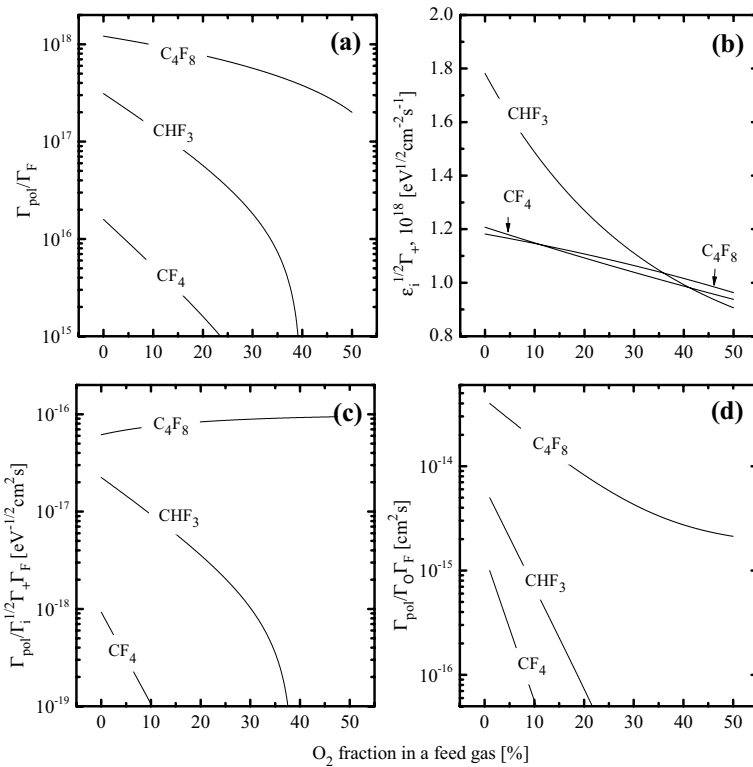
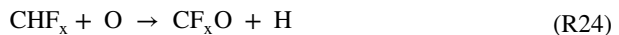


Fig. 4 Fluxes and flux-to-flux ratios characterizing etching and polymerization kinetics as functions of O_2 content in a feed gas: **a** Γ_{pol}/Γ_F ratio characterizing polymer deposition rate; **b** parameter $\sqrt{\epsilon_i}\Gamma_+$ characterizing ion bombardment intensity; **c** parameter $\Gamma_{pol}/\sqrt{\epsilon_i}\Gamma_+\Gamma_F$ characterizing the change in polymer film thickness due to the physical etching pathway; **d** parameter $\Gamma_{pol}/\Gamma_O\Gamma_F$ characterizing the change in polymer film thickness due to the chemical etching pathway. Processing conditions correspond to Fig. 1



The important feature is that the contribution of R20 to the total decay rate for F atoms exceeds those for R6 and R7. This produces the noticeably lower F atom density compared with the $CF_4 + Ar$ gas system, though the total F atom formation rate in the $CHF_3 + Ar$ plasma is a bit higher. The substitution of Ar for O_2 in the $CHF_3 + O_2 + Ar$ gas mixture retards the electron-impact dissociation kinetics (because of sufficient falls in both T_e and n_e , see Fig. 1) as well as strongly suppresses all F atom formation pathways which do work in the $CHF_3 + Ar$ plasma. The last effect is caused by a) fast losses of CF_x and CHF_x radicals (Fig. 3(b)) due to their effective conversion into CF_xO species in R8, R24 ($k_{24} \sim 1.1 \times 10^{-11} \text{ cm}^3/\text{s}$ for $x=2$) and R25 ($k_{25} \sim 3.5 \times 10^{-11} \text{ cm}^3/\text{s}$ for $x=1$):



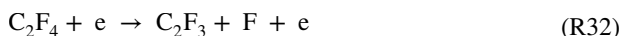
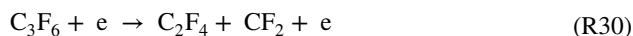
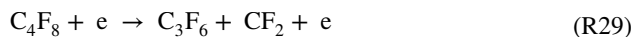


and b) the decreasing rate of R23 due to the tenfold fall in dissociative collision frequency for electrons with HF molecules ($k_{23}n_e = 88.5\text{--}8.3 \text{ s}^{-1}$ at 0–50% O_2). The other remarkable differences in respect to the previous gas system are the lower production rates for O and O^* in electron-impact reactions R26 and R27:



as well as the faster overall decay of oxygen atoms through R8, R24 and R25. As a result, the smaller amount of O and O^* in a gas phase (Fig. 3d) produces lower densities of CF_xO and FO species compared with CF_4 -based plasma (Fig. 3e) and limits the overall contribution of R9, R10 and R16–R18 to the F atom formation kinetics. That is why the total F atom formation rate decreases monotonically (by ~ 2.3 times at 0–50% O_2) toward O_2 -rich plasmas. At the same time, rapidly decreasing densities of CHF_x and CF_x radicals also reduce the decay frequency for F atoms in both heterogeneous (R7) and gas-phase (R20) reactions. The last tendency appears to be faster compared with a change in the fluorine atom formation rate and causes the monotonic increase in the F atom density, as shown in Fig. 3a. Opposite tendencies for Γ_{pol} and Γ_F provide a rapid decrease in the Γ_{pol}/Γ_F ratio toward O_2 -rich plasmas (Fig. 4a). This surely points out on decreasing polymer deposition rate and plasma polymerizing ability.

In the $\text{C}_4\text{F}_8 + \text{Ar}$ plasma, the gas phase is mostly composed by fluorocarbon components CF_x ($x = 1, 2, 3$) and C_2F_x ($x = 3, 4$) (Fig. 2c) [14–17]. These particles appear as the first-step dissociation products of original C_4F_8 molecules in R28 and R29 well as result from the further decomposition of corresponding reaction products through R5 for $x = 2$, R30, R31 and R32. The main source of F atoms is given by R5 for $x = 1\text{--}3$ while their decay in addition to R6 and R7 is noticeably contributed by R33 ($k_{33} \sim 4.0 \times 10^{-11} \text{ cm}^3/\text{s}$):



Accordingly, the domination of CF_2 radicals over other neutral components is provided by their high formation rate in R29–R31 and R33. The total density of F atoms exceeds that in the $\text{CHF}_3 + \text{Ar}$ plasma as well as has the same level with the corresponding value in $\text{CF}_4 + \text{Ar}$ plasma. The last fact is in good agreement with experimental data [35]. The substitution of Ar for O_2 in the $\text{C}_4\text{F}_8 + \text{O}_2 + \text{Ar}$ gas mixture also reduces the efficiency of R5 (due to the simultaneous decrease in T_e and n_e , see Fig. 1) as well as initiates the

decomposition of CF_x radicals through R8. At the same time, as can be seen from Fig. 3b, a decrease in n_{CF_x} toward higher y_{O_2} values appears to be much slower compared with CF_4 - and CHF_3 -based plasmas. The reason is the effective loss of O_2 molecules in R34 ($k_{34} \sim 3.2 \times 10^{-11} \text{ cm}^3/\text{s}$) and R35 ($k_{35} \sim 1.5 \times 10^{-11} \text{ cm}^3/\text{s}$):



that results in lower n_{O_2} values (Fig. 3e) and provides maximum densities for hardly dissociated CO and CO_2 species (Fig. 3f). In fact, such situation limits formation rates for O and O^* through R26 and R27 as well as causes lower O atoms densities compared with CF_4 - and CHF_3 -based plasmas (Fig. 3d). The lack of oxygen atoms reduces the significance of R8–R10 and R16–R18 in respect to production of F atoms, so that their overall contribution do not compensate for the fall of R5 even at $y_{\text{O}_2} > y_{\text{Ar}}$. Accordingly, the total F atom formation rate exhibit the monotonic decrease toward higher y_{O_2} values, and the same behavior is also for the F atoms density, as shown in Fig. 3a. Though changes in Γ_{pol} and Γ_F are qualitatively similar, the latter appears to be slower in the quantitative scale. Such situation corresponds to the weakly decreasing $\Gamma_{\text{pol}}/\Gamma_F$ ratio (Fig. 4a) that suggests the same behavior of polymer deposition rate.

Data of Figs. 3 and 4 allow one to formulate features which may influence heterogeneous process kinetics in given gas systems. First, the substitution of Ar for O_2 in all three gas mixtures introduces identical gas-phase reaction mechanisms to reduce densities of polymerizing radicals, but has specific impacts on the formation-decay balance for F atoms. As a result, an increase in y_{O_2} up to 50% does not change the basic rule concerning the correlation between the polymerizing ability and the z/x ratio in original $\text{C}_x\text{H}_y\text{F}_z$ molecules, but disturbs the difference in n_F compared with non-oxygenated mixtures. Particularly, the rapid fall of both n_F and Γ_F in the $\text{C}_4\text{F}_8 + \text{O}_2 + \text{Ar}$ plasma finally leads to lowest values of these parameters among other O_2 -rich mixtures. Another principal effect is that even the formally similar changes of F atom density in $\text{CF}_4 + \text{O}_2 + \text{Ar}$ and $\text{CHF}_3 + \text{O}_2 + \text{Ar}$ plasmas are caused by different reasons. These are changes in F atom formation or decay kinetics, respectively. Second, the substitution of Ar for O_2 always suppresses the polymer deposition rate through increasing gap between densities of F atoms and polymerizing radicals. The stronger effect for $\text{CF}_4 + \text{O}_2 + \text{Ar}$ and $\text{CHF}_3 + \text{O}_2 + \text{Ar}$ plasmas (Fig. 4a) is because of increasing density of F atoms and opposite changes in Γ_F and Γ_{pol} . And thirdly, the substitution of Ar for O_2 in all three gas systems results in decreasing polymer film thickness, h_{pol} . Again, the deeper fall of h_{pol} in $\text{CF}_4 + \text{O}_2 + \text{Ar}$ and $\text{CHF}_3 + \text{O}_2 + \text{Ar}$ plasmas is due to the simultaneous acceleration in physical (Fig. 4c) and chemical (Fig. 4d) polymer decomposition pathways. In the $\text{C}_4\text{F}_8 + \text{O}_2 + \text{Ar}$ plasma, the decrease in $\sqrt{\epsilon_i}\Gamma_+$ characterizing the ion bombardment intensity (Fig. 4b) has almost the same slope with the change in $\Gamma_{\text{pol}}/\Gamma_F$ (Fig. 2a). Such situation causes the nearly constant efficiency for sputter etching of polymer film in the range of 0–50% O_2 .

From Fig. 5a, it can be seen that the Si etching rates in in $\text{CF}_4 + \text{O}_2 + \text{Ar}$, $\text{CHF}_3 + \text{O}_2 + \text{Ar}$ and $\text{C}_4\text{F}_8 + \text{O}_2 + \text{Ar}$ plasmas are characterized by different absolute values but exhibit similar non-monotonic changes with increasing O_2 fraction in a feed gas. Experiments with both pure Ar and 50% Ar + 50% O_2 plasmas indicated that corresponding etching rates (in fact, rates of physical sputtering, R_{phys}) are quite close and do not exceed 5 nm/min. The correction of these values according to the change of $\sqrt{\epsilon_i}\Gamma_+$ in fluorocarbon-containing plasmas confirmed that the range of 0–50% O_2 surely provide the condition $R_{\text{phys}} < R_{\text{chem}}$ and $R \approx R_{\text{chem}}$. Therefore,

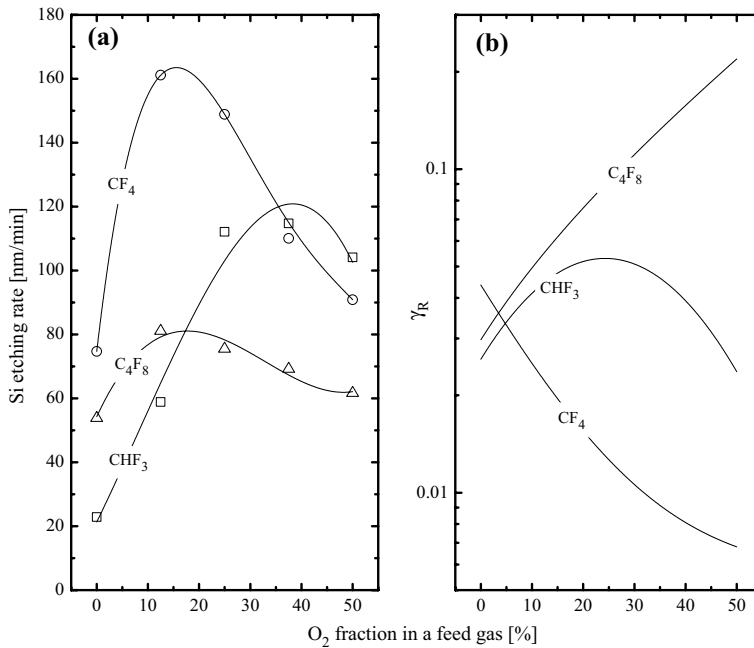


Fig. 5 Si etching rate (a) and effective reaction probability (b) as functions of O₂ content in a feed gas. Processing conditions correspond to Fig. 1

one can conclude that a) the dominant etching pathway in all three cases is the heterogeneous chemical reaction

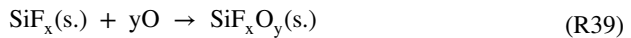


where index (s.) relate the surface-bonded state of corresponding particle; and b) the non-monotonic $R_{chem} = f(y_{O_2})$ curves in each gas system contradict with monotonic change of F atom flux. The last fact directly points out on non-constant effective reaction probabilities $\gamma_R \approx R/\Gamma_F$ which depend on the feed gas composition (Fig. 5b).

In the case of low-polymerizing CF₄+O₂+Ar plasma, the monotonically decreasing γ_R contradicts with the change of h_{pol} (since the latter also decreases toward O₂-rich plasmas, see Fig. 4c, d), but shows the formal agreement with $\sqrt{\epsilon_i}\Gamma_+$ (Fig. 4b). At the same time, the mainly spontaneous mechanism of R36 must be low sensitive to the ion bombardment intensity [5]. That is why, one can suggest that a decrease in γ_R is only partly supported by the change in $\sqrt{\epsilon_i}\Gamma_+$, but is mostly caused by heterogeneous effects with a participation of oxygen atoms. When assuming the rather thin or even the non-continuous fluorocarbon polymer film which does not influence the etching kinetics, there are at least two reasonable mechanisms which should be taken into account. The first one is the competitive adsorption of O atoms with a formation of oxide bonds R37 while the back process R38 has the ion-driven nature:



As such, one can reasonably suggest that an increase in y_{O_2} accelerates the threshold-less R37 (as follows from increasing O atom flux, see Fig. 3d) under the condition of slightly decreasing reaction yield for R38 (as follows from the change in $\sqrt{\varepsilon_i}\Gamma_+$, see Fig. 4b). As a result, the transition toward O_2 -rich plasmas reduces the amount of free adsorption sites for etchant species and suppresses γ_R . The similar effect was found to retard the Si etching rate in O_2 -rich $SF_6 + O_2$ plasmas [37, 38]. The second mechanism assumes that an increase in y_{O_2} changes the composition of reaction products toward lower volatile compounds. From Ref. [36], it can be understood that both $SiCl_xO_y$ and $SiBr_xO_y$ compounds have much lower volatility compared with $SiCl_x$ and $SiBr_x$. Though we have no confirmation that the same rule does work for the fluorine-based etching chemistry, the indirect proof is that SiF_xO_y works as a passivation layer in cryogenic etching processes [39, 40]. Therefore, even if SiF_xO_y exhibits the ability to the spontaneous desorption at nearly room temperatures [40], the corresponding desorption yield at given T_s may be lower compared with the non-oxidized SiF_x . As such, the oxidation of reaction products



anyway lowers their overall desorption rate and reduces the fraction of free adsorption sited for F atoms with increasing y_{O_2} . Moreover, one can suggest that an increase in “y” value in the SiF_xO_y molecule (in other words, the transition to more oxidized silicon with increasing O atom flux) may suppress the volatility down to the domination of the ion-stimulated desorption pathway R40:



As a result, the acceleration of R39 together with decreasing efficiency of R40 increases the residual amount of reaction products on the etched surface and thus, reduces γ_R through decreasing fraction of free adsorption sites for F atoms. It is important to note also that our model-predicted $\gamma_R \sim 0.044$ for 50% $CF_4 + 50\%$ Ar plasma is quite close to those obtained in experiments with non-fluorocarbon sources of fluorine atoms [41, 42]. In our opinion, this confirms the conclusion on the polymer-film-independent etching regime, at least within the given set of processing conditions.

In the case of high-polymerizing $C_4F_8 + O_2 + Ar$ plasma, the monotonic increase in γ_R contradicts with $\sqrt{\varepsilon_i}\Gamma_+$, but shows an agreement with the change in the polymer film thickness. Probably, even the full substitution of Ar for O_2 still keep the thick polymer film and thus, does not change the process limiting stage compared with the oxygen-less $C_4F_8 + Ar$ plasma. Accordingly, the decreasing h_{pol} with higher O_2 fractions in a feed gas accelerates the chemical etching pathway by providing the better access of F atoms to the etched surface (in fact, by increasing the Γ'_F/Γ_F ratio).

In the case of $CHF_3 + O_2 + Ar$ plasma, one can speak about an intermediate case when an increase in y_{O_2} provides the transition from thick to thin polymer film. It can be suggested that, under the condition of Ar-rich plasma ($y_{O_2} < y_{Ar}$), the polymer film is still thick enough to limit the rate of chemical reaction through the transport of F atoms to the etched surface. That is why the increasing tendency of γ_R up to 25–30% O_2 for all three materials correlates with the change in h_{pol} . Oppositely, the O_2 -rich plasma ($y_{O_2} > y_{Ar}$) provides the thin polymer film that satisfies the condition of $\Gamma'_F/\Gamma_F \approx 1$. As such, the further decrease in the amount of residual polymer does not influence both γ_R and kinetics of chemical etching pathway. At the same time, such situation lowers the amount of O atoms consumed for the reaction with the fluorocarbon polymer as well as increases the O atom flux reaching the etched surface. Probably, this stimulates heterogeneous reaction pathways R37–R40 as well

as lowers the effective reaction probability for F atoms through the decreasing fraction of free adsorption sites. That is why the $\gamma_R = f(y_{O_2})$ curve exhibits the non-monotonic shape, and the bend point at about $y_{O_2} \approx y_{Ar}$ reflects the change in the process limiting stage.

Since we did not provide etching experiments for SiO_2 and did not control etched profiles, we cannot characterize our gas systems in terms of etching anisotropy and SiO_2/Si etching selectivity. Instead, some reasonable suggestions in comparison with existing experimental data can be made. In particular, as follows from Ref. [20], Si and SiO_2 etching kinetics in pure CHF_3 and C_4F_8 plasmas is mainly controlled by the polymer film thickness. From Fig. 4, it can be understood that the C_4F_8 -based plasma always provides higher polymer deposition rate that allows one to assume thicker polymer films with a bigger difference between Si and SiO_2 . The last suggestion is in good agreement with higher SiO_2/Si etching selectivity obtained in Ref. [20] by etching experiments. In addition, Li et. al [23] measured a decrease in polymer deposition rate and SiO_2/Si etching selectivity with increasing O_2 fraction in the binary $C_4F_8 + O_2$ gas system. Obviously, since we obtained the similar effect for the deposition rate in $C_4F_8 + O_2 + Ar$ plasma, the similar change in etching selectivity is surely expected. One can expect also that the higher polymerizing ability of C_4F_8 -based plasma results in advanced etching anisotropy for Si due to the better passivation of sidewalls. As for CF_4 - and CHF_3 -based plasmas, their comparison is more difficult due to the lack of joint experimental studies. For instance, our model-predicted data on relative polymer deposition rates and film thicknesses suggest the higher SiO_2/Si etching selectivity for the second gas system. The direct confirmation of this fact may be found in Ref. [43]. Also, Ref. [44] reported that $CHF_3 + O_2$ plasma is featured by higher absolute etching rate and lower SiO_2/Si etching selectivity compared with the $CHF_3 + Ar$ one under one and the same operating conditions. These results generally correlate with increasing Si etching rate (Fig. 5a) and decreasing polymer deposition rate (Fig. 4a) with the substitution of Ar for O_2 in the $CHF_3 + O_2 + Ar$ plasma. As such, a decrease in SiO_2/Si etching selectivity toward O_2 -rich $CHF_3 + O_2 + Ar$ plasmas seems to be quite real.

Finally, we would like to note that the above analysis of etching kinetics and thus, the interpretation of etching mechanisms mainly works in a qualitative scale due to evident simplifications in initial assumptions. At the same time, such simplified analysis allows one to trace relationships between gas-phase and heterogeneous kinetics as well as provides quite reasonable explanations of etching experiments.

Conclusions

In this work, we investigated how the O_2/Ar ratio in $CF_4 + O_2 + Ar$, $CHF_3 + O_2 + Ar$ and $C_4F_8 + O_2 + Ar$ gas mixtures does influence plasma parameters, steady-state gas phase compositions and Si etching kinetics. It was shown that the transition toward O_2 -rich plasmas (a) causes similar changes in electrons- and ions-related plasma parameters (electron temperature, plasma density, ion bombardment energy); (b) always suppresses densities of polymerizing radicals and reduces the polymer film thickness; and (c) has the different impact on F atoms kinetics. As such, even the formally similar changes in F atom density for $CF_4 + O_2 + Ar$ and $CHF_3 + O_2 + Ar$ plasmas results from different plasma chemical reaction mechanisms. It was found that the dominant etching mechanism in all cases is represented by the chemical etching pathway while the latter does not follow the behavior of F atom flux due to the process-condition-dependent effective reaction probability for F atoms. The analysis of correlations between effective reaction probability and fluxes (or

flux-to-flux ratios) of plasma active species allowed one to suggest the limiting stage of Si etching process. Depending on gas system, this may be either the transport of F atoms through thick polymer film or heterogeneous reactions with a participation of oxygen atoms under the condition of thin polymer film.

Acknowledgements This work was supported by the Industrial Technology Innovation Program (20009818, Development of plasma realtime process monitoring sensor and process simulation technology) funded by the Ministry of Trade, Industry and Energy (MOTIE, Republic of Korea) (N. Lim and K.-H. Kwon).

References

1. Sze SM (1988) VLSI technology. McGraw-Hill, New York
2. Sugano T (1990) Applications of plasma processes to VLSI technology. Wiley, New York
3. Rooth JR (1995) Industrial plasma engineering. IOP Publishing LTD, Philadelphia
4. Wlof S, Tauber RN (2000) Silicon processing for the VLSI era. Volume 1. Process technology. Lattice Press, New York
5. Lieberman MA, Lichtenberg AJ (1994) Principles of plasma discharges and materials processing. Wiley, New York
6. Roosmalen AJ, Baggerman JAG, Brader SJH (1991) Dry etching for VLSI. Plenum Press, New-York
7. Kimura T, Noto M (2006) Experimental study and global model of inductively coupled CF_4/O_2 discharges. *J Appl Phys* 100:063303
8. Kimura T, Ohe K (1999) Probe measurements and global model of inductively coupled Ar/ CF_4 discharges. *Plasma Sour Sci Technol* 8:553
9. Chun I, Efremov A, Yeom GY, Kwon K-H (2015) A comparative study of $\text{CF}_4/\text{O}_2/\text{Ar}$ and $\text{C}_4\text{F}_8/\text{O}_2/\text{Ar}$ plasmas for dry etching applications. *Thin Solid Films* 579:136
10. Efremov A, Lee J, Kim J (2017) On the control of plasma parameters and active species kinetics in $\text{CF}_4 + \text{O}_2 + \text{Ar}$ gas mixture by CF_4/O_2 and O_2/Ar mixing ratios. *Plasma Chem Plasma Process* 37:1445
11. Ho P, Johannes JE, Buss RJ (2001) Modeling the plasma chemistry of C_2F_6 and CHF_3 etching of silicon dioxide, with comparisons to etch rate and diagnostic data. *J Vac Sci Technol B* 19:2344
12. Efremov AM, Murin DB, Kwon K-H (2018) Parameters of plasma and kinetics of active particles in $\text{CF}_4(\text{CHF}_3) + \text{Ar}$ mixtures of a variable initial composition. *Russ Microelectron* 47:371
13. Efremov AM, Murin DB, Kwon K-H (2020) Plasma parameters and kinetics of active particles in the mixture $\text{CHF}_3 + \text{O}_2 + \text{Ar}$. *Russ Microelectron* 49:233
14. Kokkoris G, Goodyear A, Cooke M, Gogolides E (2008) A global model for C_4F_8 plasmas coupling gas phase and wall surface reaction kinetics. *J Phys D Appl Phys* 41:195211
15. Rauf S, Ventzek PL (2002) Model for an inductively coupled Ar/c- C_4F_8 plasma discharge. *J Vac Sci Technol A* 20:14
16. Lim N, Efremov A, Kwon K-H (2019) Gas-phase chemistry and etching mechanism of SiN_x thin films in $\text{C}_4\text{F}_8 + \text{Ar}$ inductively coupled plasma. *Thin Solid Films* 685:97–110
17. Lee BJ, Efremov A, Nam Y, Kwon K-H (2020) Plasma parameters and silicon etching kinetics in $\text{C}_4\text{F}_8 + \text{O}_2 + \text{Ar}$ gas mixture: effect of component mixing ratios. *Plasma Chem Plasma Process* 40:1365–1380
18. Schaepekens M, Standaert TEFM, Rueger NR, Sebel PGM, Oehrlein GS, Cook JM (1999) Study of the SiO_2 -to- Si_3N_4 etch selectivity mechanism in inductively coupled fluorocarbon plasmas and a comparison with the SiO_2 -to-Si mechanism. *J Vac Sci Technol A* 17:26
19. Matsui M, Tatsumi T, Sekine M (2001) Relationship of etch reaction and reactive species flux in $\text{C}_4\text{F}_8/\text{Ar}/\text{O}_2$ plasma for SiO_2 selective etching over Si and Si_3N_4 . *J Vac Sci Technol A* 19:2089
20. Standaert TEFM, Hedlund C, Joseph EA, Oehrlein GS, Dalton TJ (2004) Role of fluorocarbon film formation in the etching of silicon, silicon dioxide, silicon nitride, and amorphous hydrogenated silicon carbide. *J Vac Sci Technol A* 22:53
21. Kastenmeier BEE, Matsuo PJ, Oehrlein GS (1999) Highly selective etching of silicon nitride over silicon and silicon dioxide. *J Vac Sci Technol A* 17:3179
22. Lele C, Liang Z, Linda X, Dongxia L, Hui C, Tod P (2009) *J Semicond* 30:033005
23. Li X, Ling L, Hua X, Fukasawa M, Oehrlein GS, Barela M, Anderson HM (2003) Effects of Ar and O_2 additives on SiO_2 etching in C_4F_8 -based plasmas. *J Vac Sci Technol A* 21:284

24. Son J, Efremov A, Chun I, Yeom GY, Kwon K-H (2014) On the LPCVD-formed SiO₂ etching mechanism in CF₄/Ar/O₂ inductively coupled plasmas: effects of gas mixing ratios and gas pressure. *Plasma Chem Plasma Process* 34:239
25. Lee J, Efremov A, Yeom GY, Lim N, Kwon K-H (2015) Application of Si and SiO₂ etching mechanisms in CF₄/C₄F₈/Ar inductively coupled plasmas for nanoscale patterns. *J Nanosci Nanotechnol* 15:8340
26. Efremov A, Murin D, Kwon K-H (2020) Concerning the effect of type of fluorocarbon gas on the output characteristics of the reactive-ion etching process. *Russ Microelectron* 49:157
27. Shun'ko EV, (2008) *Langmuir probe in theory and practice*. Universal Publishers, Boca Raton
28. Vasenkov AV, Li X, Oehlein GS, Kushner MJ (2004) Properties of c-C₄F₈ inductively coupled plasmas. II. Plasma chemistry and reaction mechanism for modeling of Ar/c-C₄F₈/O₂ discharges. *J Vac Sci Technol A* 22:511
29. Lee C, Lieberman MA (1995) Global model of Ar, O₂, Cl₂, and Ar/O₂ high density plasma discharges. *J Vac Sci Technol A* 13:368
30. Hsu CC, Nierode MA, Coburn JW, Graves DB (2006) Comparison of model and experiment for Ar, Ar/O₂ and Ar/O₂/Cl₂ inductively coupled plasmas. *J Phys D Appl Phys* 39:3272
31. Chistophorou LG, Olthoff JK (2004) *Fundamental electron interactions with plasma processing gases*. Springer, New York
32. Efremov A, Lee J, Kwon K-H (2017) A comparative study of CF₄, Cl₂ and HBr + Ar inductively coupled plasmas for dry etching applications. *Thin Solid Films* 629:39
33. Proshina O, Rakhimova TV, Zotovich A, Lopaev DV, Zyryanov SM, Rakhimov AT (2017) Multifold study of volume plasma chemistry in Ar/CF₄ and Ar/CHF₃ CCP discharges. *Plasma Sources Sci Technol* 26:075005
34. Takahashi K, Hori M, Goto T (1994) Characteristics of fluorocarbon radicals and CHF₃ molecule in CHF₃ electron cyclotron resonance downstream plasma. *Jpn J Appl Phys* 33:4745
35. Sasaki K, Kawai Y, Kadota K (1999) Determination of fluorine atom density in reactive plasmas by vacuum ultraviolet absorption spectroscopy at 95.85 nm. *Rev Sci Instrum* 70:76–81
36. Kim DK, Kim YK, Lee H (2007) A study of the role of HBr and oxygen on the etch selectivity and the post-etch profile in a polysilicon/oxide etch using HBr/O₂ based high density plasma for advanced DRAMs. *Mater Sci Semicond Process* 10:41
37. d'Agostino R, Flamm DL (1981) Plasma etching of Si and SiO₂ in SF₆-O₂ mixtures. *J Appl Phys* 52:162
38. Knizikevicius R (2010) Simulations of Si and SiO₂ Etching in SF₆ + O₂ Plasma. *Acta Phys Pol A* 117:478
39. Mellhaoui X, Dussarta R, Tillocher T, Lefaucheux P, Ranson P (2005) SiO_xF_y passivation layer in silicon cryoetching. *J Appl Phys* 98:104901
40. Pereira J, Pichon LE, Dussart R, Cardinaud C, Duluard CY, Oubensaid EH, Lefaucheux P, Boufnichel M, Ranson P (2009) In situ x-ray photoelectron spectroscopy analysis of SiO_xF_y passivation layer obtained in a SF₆/O₂ cryoetching process. *Appl Phys Lett* 94:071501
41. Mullins CB, Coburn JW (1994) Ion-beam-assisted etching of Si with fluorine at low temperatures. *J Appl Phys* 76:7562
42. Chevolleau T, Tessier PY, Cardinaud C, Turban G (1997) Etching of Si at low temperatures using a SF₆ reactive ion beam: Effect of the ion energy and current density. *J Vac Sci Technol A* 15:2661
43. Yeom GY, Kushner MJ (1990) Si/SiO₂ etch properties using CF₄ and CHF₃ in radio frequency cylindrical magnetron discharges. *Appl Phys Lett* 56:857
44. Gatzert C, Blakers AW, Deenapanray PNK, Macdonald D, Auret FD (2006) Investigation of reactive ion etching of dielectrics and Si in CHF₃/O₂ or CHF₃/Ar for photovoltaic applications. *J Vac Sci Technol A* 24:1857